6. Automated and emerging technologies

6.2 Robotics

Marking scheme

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Q1)

Question	Answer	Marks
(a)	Any **two** from: • It has a **mechanical** structure/framework • It has **electrical** components // by example	2
(b)	Any **two** from: e.g. • Employees don't need to lift heavy furniture • Employees can be protected from dangerous tasks • Employees can utilise their skills in other tasks • Employees don't need to perform repetitive/mundane tasks	2

Question	Answer	Marks
(c)	Any **one** from: e.g. • Expensive to **install/purchase/setup** • High ongoing costs/maintenance costs • May deskill the workforce • If they malfunction, production may stop	1

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Q2)

Question	Answer	Marks
(a)	The robot can perform actions **without human intervention**	1
(b)	Any **three** from: • It has a **mechanical structure/framework** • It has **electrical** components • It is programmable • It can move	3
(d)	Any **two** from: Example: • The robot is more accurate when planting seeds • The robot does not need to take breaks // works 24/7 • The robot will not get bored of performing a repetitive task • The farmer can perform other tasks whilst the robot is planting the seeds • The robot may be faster/more efficient (at planting seeds) • Safer as the robot can work in a more dangerous environments (e.g. with animals in a field) • Doesn't need to pay **labour** costs/**wages**	2

Question	Answer	Marks
(e)	Any **two** from: Example: • The robot may have been expensive to **purchase** • The robot may need **maintenance** and this may be costly • The robot may take away jobs from humans that previously planted the seeds • Farmer/workers may be deskilled • Farmer may need training in how to use the robot • If it breaks the farmer will need to manually plant • If faulty it may cause damage to farm/seeds • Cannot adapt to a **new/unexpected** event	2
(f)	Any **two** from: Example: • The robot can now **adapt itself** by changing its own **rules/data/ processes** • The robot will become more efficient • … as it has a **greater knowledge** of its **surroundings/environment** • … as it can remember where a fence is • … as it can remember the route it needs to take • … as it can remember where any obstacles are to avoid • … as it can remember where to start and stop sowing seeds • … as it may make fewer errors	2

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6. Automated and emerging technologies

6.3 Artificial intelligence

Marking scheme

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Q1)

Question	Answer	Marks
(a)	**Three** from: • Rule base • Knowledge base • Interface	3
(b)	Any **two** from: • It makes decisions • … by **applying** the rules/logic to the facts/knowledge … • … to provide a result/diagnosis	2

Q2)

Question	Answer	Marks
(a)	• Interface • Knowledge base	2
(b)	Any **two** from: • **Stores** the rules for the system • … for the inference engine to use • Used to link the facts in the knowledge base	2

Q3)

Question	Answer	Marks
(a)	Any **one** from: – The ability to learn/adapt // machine learning abilities – The collection of data and the rules for using that data – The ability to reason // has problem solving abilities // makes predictions – Simulates intelligent/human behaviour – Analyses patterns	1
(b)	Any **six** from: – It has an interface … – … used to input data/view output – It has a knowledge base – It has a rule base – It has an inference engine – Applies the rule base to/and the knowledge base to provide output/diagnosis/result/solution/decision – Decides what to ask next based on the data input	6

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Q4)

Question	Answer	Marks
(a)	**One** mark for each correct term, in the correct order: – Knowledge base – Inference engine – Rule base // knowledge base – Knowledge base // rule base – Interface	**5**

Question	Answer	Marks
(b)	Any **four** from: e.g. – It is a form of artificial intelligence – Means it can adapt/change (its own processes) // It can edit its own algorithms – It can edit its own data – It can be trained – … this can be supervised/unsupervised – … meaning it can learn with/without human interaction – Analyses patterns and stores **successful/unsuccessful** results … – … to influence future decisions – (Supervised) means a user tells the system the input and output – (Unsupervised) means the system is given the input and needs to work out the output	**4**

Q5)

Question	Answer	Marks
(a)	Any **two** from: • The **simulation** of intelligent behaviours by computers • A collection of data and the rules for using that data • Has the ability to reason • Has the ability to learn/adapt	**2**
(b)	Any **five** from: • The user will enter data into the **interface** // The user will plug the car into an **interface** // The user is given questions using the **interface** • The **inference engine** will decide which questions to ask • … by using the previous answers given • The **inference engine** will decide on a diagnosis • … by **comparing the data/answers** entered to the **knowledge base** and **rule base** • … by calculating which option is most likely if there are multiple • The **interface** will output the **diagnosis/result** • The **explanation system** shows **how** the diagnosis was reached	**5**

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Q6)

Question	Answer	Marks
(a)	**One** mark for each correct term in the correct order. • artificial • interface • inference engine • knowledge base • rule base • inference engine • interface	7
(b)	Machine learning	1

Q7)

(b)	• Knowledge base • Rule base • Inference engine	3
(c)	Any **three** from: For example: • It could **gather data** during **vacuuming** • … and adapt its **own processes** • … such as where obstacles are placed in the room • … such as where dirtier areas are • … such as a path through a room • … such as the shape of a room • … such as the most efficient route to vacuum a room • … so, it knows areas to avoid/concentrate on/use different cleaning tools	3

Q8)

(d)	Any **four** from: For example: • It could make use of machine learning • It could **gather data** from **ploughing** • … and use this data to **adapt** its **own process** • … so that it will make fewer mistakes • … such as the dimensions of the field • … such as the landscape of the field • … such as where obstacles are in the field • … to create a map of the field • … to develop the most efficient route to take • … so, it knows what to **avoid** in future	4

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